

Unsupervised Anomaly Detection Using K-Means, Local Outlier Factor and One Class SVM

Efrem Heri Budiarto
*Department of Electrical Engineering
 and Information Technology
 Universitas Gadjah Mada
 Yogyakarta, Indonesia
 efremheribudiarto@mail.ugm.ac.id*

Adhistya Erna Permanasari*
*Department of Electrical Engineering
 and Information Technology
 Universitas Gadjah Mada
 Yogyakarta, Indonesia
 adhistya@ugm.ac.id*

Silmi Fauziati
*Department of Electrical Engineering
 and Information Technology
 Universitas Gadjah Mada
 Yogyakarta, Indonesia
 silmi@ugm.ac.id*

Abstract—Anomaly detection is one area that is still being researched today. An anomaly that occurs in data can be utilized in various ways, such as detection of money embezzlement, increasing production, improving the quality of data, preventing attacks on a network and others. This study aims to find anomalies in the data drug use in hospitals consisting of two datasets using the K-Means algorithm, Local Outlier Factor (LOF) and One Class Support Vector Machine (OC-SVM). The results of this study are that the three algorithms can find outliers. Based on its performance in both datasets, OC-SVM outperforms LOF and K-Means

Keywords—outlier detection, anomaly detection, one class SVM, k-means, local outlier factor, data mining.

I. INTRODUCTION

Anomaly detection or commonly known as outlier detection can be applied to various fields such as banking, healthcare, networks, capital markets, and more. Because of that, anomaly detection has a different meaning depending on the field that implements it [1]. For example, in banking this anomaly is used to find irregularities in customer transactions; on health anomalies can be used to detect irregularities in EEG results, ECG even in certain image analyzes; on networks commonly used to look for suspicious activities from network users that are destructive or annoying to other users; in the field of capital markets can be used to find certain patterns that can have a negative impact on market conditions. In the statistics and data mining fields anomaly can make the results of data analysis biased, so the results cannot describe the actual phenomenon. Another impact is that it can reduce accuracy in making predictions.

Anomaly detection has been studied since first using statistical techniques, data mining, and machine learning. Technological developments also make data sources to carry out these tests more diverse such as text, images, digital documents, data streaming and more. Anomaly detection can also be distinguished based on datasets, such as Time-series vs. Multidimensional Data Facet, The Point vs. Window Facet, The Data Type Facet, and Facet supervision [2].

This study aims to find anomalies in the data and compare the time of execution. The type of data used is time-series which consists of 2 datasets of drug use every week [3]. The techniques chosen are K-Means, One Class Support vector machines (OC-SVM), and Local Outlier

Factor (LOF). This method is commonly used to find outliers in the case of an unlabeled dataset by grouping the data based on the distance between the data [2].

II. RELATED WORKS

Several studies have been performed using several algorithms in various fields. Kumari *et al.* [4] conduct research related traffic anomaly detection on the network by using the K-Means and Big Data as a data source. This is done to find suspicious behavior from users that is destructive. The research was conducted by grouping user activities with certain codes, this can prevent attacks/damage to certain networks. Zhang *et al.* [5] research related to the detection of the condition of the battery by determining the battery characteristics such as the status of charging, discharging, and released. The method used to determine whether a battery is still in good condition or not is by comparing these characteristics in the form of a certain curve to the results of anomaly estimates using the K-Means method. Through this research, the battery condition can be known in a more reliable, safe and efficient way. Sarvani *et al.* [6] used K-Means to eliminate outliers in data. Data with the farthest distance from the centroid is called an outlier.

Fouzi *et al.* [7] used One Class SVM to detect anomaly in a photovoltaic solar power plant. The research aims to enable the plant to work safely, reduce operating costs, increase operating profits and be reliable. SVM method is used to measure the difference between normal and abnormal features, this is supported by SVM capabilities that can handle nonlinear data. Yahaya *et al.* [8] conducted research related to the Activities of Daily Living (ADL) which focused on one's sleep patterns. Through SVM, datasets that have not been labeled can be labeled 1 for data that is considered normal and -1 for abnormal. Devi *et al.* [9] utilized SVM combined with the Tomek-link method to handle cases of class imbalance by adopting the concept of anomaly detection.

The algorithm commonly used next is LOF, Mishra *et al.* [10] use it to detect anomalies in data streams. The research focuses on density-based outliers between data and considers performance based on certain parameters. Huang said [11] the conducted research utilizing LOF and K Nearest Neighbor (K-NN) to detect anomalous behavior in ships. This research uses KNN to filter existing vessel datasets, then the data is calculated by the local deviation index using the LOF algorithm, the index results are

compared with a certain threshold so that it can be determined normally or not. Such research can assist in monitoring ship safety.

III. RESEARCH METHODOLOGY

This study uses a dataset of drug use for 1 year in hospitals [3]. There are 2 datasets, first Cephalaxine drugs with 52 data and Enxyples drugs with 50 data. We use Google Colab as a tool to test the dataset. Google Colab is a tool made by Google to help researchers in processing data for learning and experimentation, especially in the field of Machine Learning. Basically, this tool is a Jupyter Notebook that uses the Python language as its programming language. Some libraries are used in this study such as numpy, pandas, matplotlib, sklearn, memory profiler and scipy.

We use the 0.01 and 0.05 thresholds as the lower limit of this research on anomaly detection. The limit is used based on the possibility of anomalous values in the 3rd quartile on normally distributed data.

A. Dataset Attributes

The dataset used in this research is data on drug use per week, with attributes such as the following:

- 1) *Week*: Number of the week.
- 2) *N_Medicine*: Amount of drug use in one week.

Below is the distribution of data for each dataset:

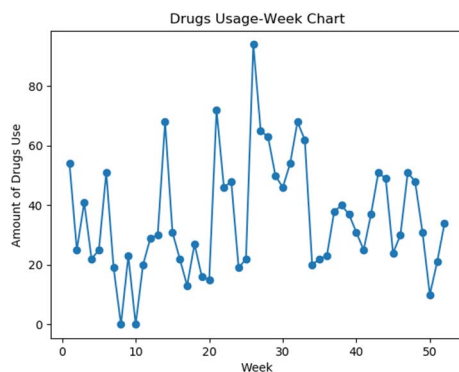


Fig.1. Data graph distribution for dataset1

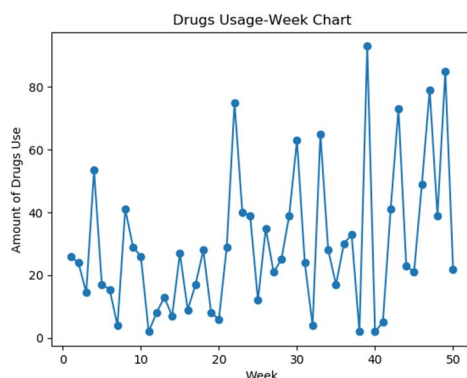


Fig.2. Data graph distribution for dataset2

Figure 1 and Figure 2 illustrate the distribution between the number of drugs (*n_medicine*) which are on the Y-axis and week (*week*) on the X-axis. Based on the picture it can be seen that the week does not have too much influence on drug use. Another way that can be used to find out the influence between these attributes is to do a

correlation test. Based on the correlation test between attributes the results for each dataset are 0.129 and 0.353 respectively. This proves that the column for drug use does not depend on the week column.

B. Preprocessing

In the preprocessing stage, we check the missing values in each dataset. Figure 1 and Figure 2 explain that there is no missing value, so there is no need to take special action. Next, we normalize using MinMax Scaler. This needs to be done so that the existing datasets can be used into a predetermined algorithm.

C. K-Means

K-Means is a clustering algorithm that can group data by comparing its mean value [12]. For this algorithm to work, the number of clusters needs to be determined first. This method can be denoted as follows using Equation (1) and (2):

1) Assignment Step

$$S_i^{(t)} = \{x_p : \|x_p - m_i\|^2 \leq \|x_p - m_j\|^2 \forall j, 1 \leq j \leq k\} \quad (1)$$

2) Update Step

$$m^{(t+1)} = \frac{1}{|S_i^{(t)}|} \sum_{x_j \in S_i^{(t)}} x_j \quad (2)$$

This algorithm can find outliers by grouping data into a cluster. Then each cluster center point will be compared with all its members. Members who have a certain distance can be considered as an anomaly.

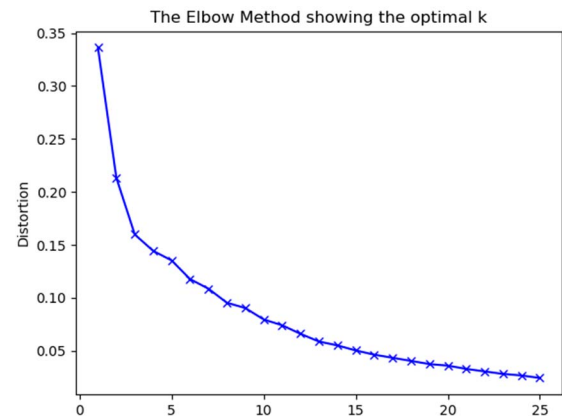


Fig.3. Elbow method test for dataset1

This research uses the elbow method for each dataset to determine the number of clusters. The elbow method describes how the algorithm performs by comparing the value of the Sum of Square Error (SSE) on each loop [13]. Figure 3 illustrates how the performance of the first dataset when tested with the elbow method with 25 repetitions. Based on the test determined the cluster value used is 5 in the first dataset and 6 in the second dataset.

D. One Class SVM

One Class SVM (OC-SVM) is a technique that was developed from the SVM. SVM is an algorithm to classify a dataset. In simple terms, SVM works by finding the best hyperplane to separate data. Based on this concept, OC-SVM can be used to find outliers in a dataset based on a

certain threshold. The output of this algorithm is data with labels normal (+) and not (-) [14].

E. Local Outlier Factor

Local Outlier Factor is an algorithm that is used to find outliers on data effectively by comparing the density of each data [11] as in Figure 4. Determining the anomaly in a data is done by calculating the deviation of data against another data, the lowest density is labeled abnormally. The output of this algorithm is data labeled 1 for normal data and -1 for anomalies.

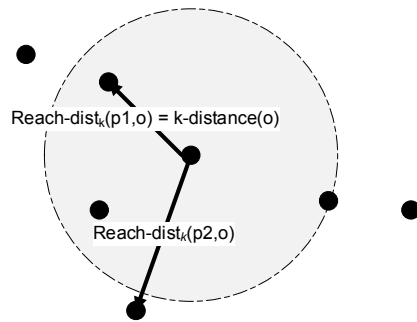


Fig.4. How LOF Algorithm Works [15]

F. Experimental Metrics

The determination of experimental metrics is not an easy thing, especially in the case of unlabeled datasets. In research [4][5][8][9] aims to find errors or irregularities in data. The metrics used to compare the algorithms in this study are the average computational speed in units of a microsecond (μ s), the average memory performance in units of MegaByte (MB) and the ability to find anomalies in the data. Computational averages and memory performance are calculated by looping ten times.

IV. RESULTS AND DISCUSSION

In this section, we compare the performance of each algorithm based on parameters and predetermined thresholds. This test is done using Google Colab. The experimental metrics used are CPU time in units of a microsecond (μ s), memory performance in units of MegaByte (MB) and the number of anomalies found in the data. The selection of the unit is because the unit is a commonly used unit.

The test results of each dataset can be seen in Table I and Table II. The table head is given a description of the method name, parameters, number of anomalies found, average computational speed in units of μ s or equal to 0.000001 seconds and average memory usage for algorithm extrusion in units Megabyte (MB).

TABLE I. DATASET1 RESULT

Method	Parameter	Anomaly	CPU Time (μ s)	Peak Memory (MB)
K-Means5	cluster = 5 outlier_limit= 0.05	2 (index 49, 50)	3.3	164

K-Means1	cluster=5 outlier_limit= 0.01	0	7.1	164.06
LOF5	contamination n=0.05	2 (index 20, 25)	3.2	164
LOF1	contamination n=0.01	1 (index 25)	3.5	164.08
OC-SVM5	nu = 0.05, gamma=auto	3 (index 0, 7, 25)	3	164.01
OC-SVM1	nu = 0.01, gamma=auto	2 (index 7, 25)	3.2	164.13

TABLE II. DATASET2 RESULT

Method	Parameter	Anomaly	CPU Time (μ s)	Peak Memory (MB)
K-Means5	cluster = 6 outlier_limit=0.05	2 (index 3,6)	3	162.94
K-Means1	cluster=6 outlier_limit=0.01	0	2.9	162.94
LOF5	contamination= 0.05	2 (index 38, 48)	2.8	162.95
LOF1	contamination= 0.01	0	2.8	162.95
OC-SVM5	nu = 0.05, gamma=auto	3 (index 6, 38, 48)	2.7	162.95
OC-SVM1	nu = 0.01, gamma=auto	2 (index2,6)	2.7	162.95

Based on Table I and Table II the use of 0.01 and 0.05 thresholds has an impact on the sensitivity of the algorithm. The wider the threshold, the more anomalies in the dataset are found. On the other hand, CPU Time and Peak Memory have little impact on changes in threshold values. The difference between the metrics is also not too far away.

The slight differences in CPU Time between each algorithm makes it difficult to draw a conclusion. So it is necessary to do a significance test. This study used the Kruskal Wallis H-test to determine the CPU Time difference between groups. The test does not require normally distributed data. The confidence level used in this test is 0.05. Kruskal Wallis H-test gives a significance value of 0.456 and 0.685 in each dataset. Based on these results we know that there are no significant differences between CPU Time of each algorithm.

An interesting thing was found in the number of anomalies found. In the LOF and OC-SVM algorithms that indeed function to find anomalies in a dataset have results that tend to be the same. As LOF1 and OC-SVM1 in both datasets having similar results, only OC-SVM1 can find more anomalies. An interesting result is also shown by the K-Means algorithm which is not a special algorithm to find anomalies but has results that are not much different in terms of numbers.

Anomalies found by the OC-SVM algorithm and LOF tend to be similar when viewed from the order, but the K-Means results are different. The difference in anomaly can be seen in Table I, K-Means5 gets anomalies in the order of 49 and 50 while those of LOF and OC-SVM tend to find anomalies in the middle of the data. This is because

the method of determining the anomaly in K-Means compares the distance of a data with the cluster point, if the data is included in a certain threshold this will read anomalies. LOF uses a slightly different way of working, this algorithm forms clusters for each data. Determination of anomalies or not is seen whether 2 clusters can intersect, otherwise, it is referred to as an outlier.

The visualization of the anomaly in the first dataset and the second dataset can be seen in Fig.5 and Fig.6. The K-Means result is represented by a circle, a LOF with a cross and OC-SVM with a triangle. There are overlapping points as in the first dataset LOF1, LOF5, OC-SVM1 and OC-SVM5 at point 25 and K-Means5, OC-SVM1 on the second dataset at point 6.

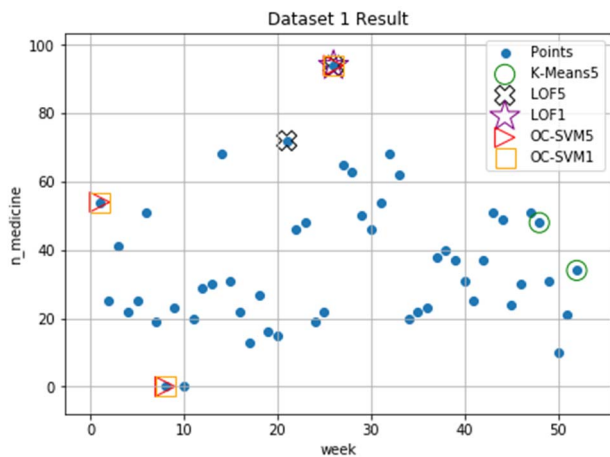


Fig.5. Dataset 1 Result Visualization

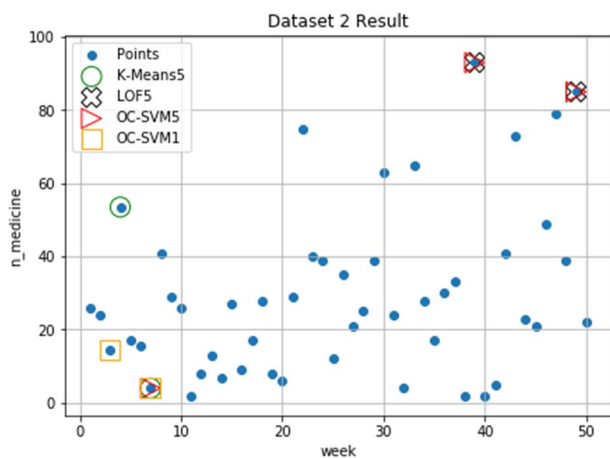


Fig.6. Dataset 2 Result Visualization

V. CONCLUSION

This study presents a comparison between K-Means, LOF, and OC-SVM. The algorithm can work with fast computing, but the right threshold is needed to get the right results. Based on this study OC-SVM with an anomalous threshold of 0.05 obtained the best results in anomalous findings on dataset 1. That algorithm found three anomalous data and the average CPU time is 0.2 μ s better than LOF. While dataset 2 with the same threshold shows OC-SVM is better at finding anomalies, on the other hand, LOF is better in performance. Based on the

CPU significance test there is no significant difference in each algorithm.

To improve the performance of each algorithm used in this study can be done by changing other parameters. For future work, this anomaly can be given certain treatments so that the quality of the data increases and the results of this study also need to be compared with other datasets.

ACKNOWLEDGMENT

The authors would like to acknowledge funding from DIKTI through the PDUPT program in 2019, Universitas Gadjah Mada.

REFERENCES

- [1] B. Hussain, Q. Du and P. Ren, "Semi-Supervised Learning Based Big Data-Driven," *China Communications*, vol. 15, no. 4, 2018, pp. 41 - 57
- [2] M. Gupta, J. Gao, C. C. Aggarwal, and J. Han, "Outlier Detection for Temporal Data: A Survey," in *IEEE Transactions on Knowledge and Data Engineering*, vol. 26, no. 9, Sept. 2014, pp. 2250-2267.
- [3] H. Pratyaksa, A. E. Permanasari, and S. Fauziati, "Application of Whittaker-Henderson Refinement Method To Improve the Accuracy of Neural Network Prediction for Predicting the Amount of Drug Use [In Bahasa Indonesia]," Universitas Gadjah Mada, Yogyakarta, 2017.
- [4] R. Kumari, Sheetanshu, M. K. Singh, R. Jha, and N. K. Singh, "Anomaly detection in network traffic using K-mean clustering," *2016 3rd International Conference on Recent Advances in Information Technology (RAIT)*, Dhanbad, 2016, pp. 387-393.
- [5] J. Zhang, L. Geng, and Y. Ma, "Battery anomaly and degradation diagnosis for renewable energy plant," in *CIREP - Open Access Proceedings Journal*, vol. 2017, no. 1, 2017, pp. 46-49.
- [6] Sarvani A., Venugopal B., Devarakonda N, Anomaly Detection Using K-means Approach and Outliers Detection Technique. In: Ray K., Sharma T., Rawat S., Saini R., Bandyopadhyay A. (eds) *Soft Computing: Theories and Applications. Advances in Intelligent Systems and Computing*, vol 742. Springer, Singapore, 2019.
- [7] Fouzi Harrou, Abdelkader Dairi, Bilal Taghezouit, Ying Sun, An unsupervised monitoring procedure for detecting anomalies in photovoltaic systems using a one-class Support Vector Machine, *Solar Energy*, Volume 179, 2019, pp. 48-58, ISSN 0038-092X, <https://doi.org/10.1016/j.solener.2018.12.045>.
- [8] Yahaya S.W., Langensiepen C., Lotfi A., Anomaly Detection in Activities of Daily Living Using One-Class Support Vector Machine. In: Lotfi A., Bouchachia H., Gegov A., Langensiepen C., McGinnity M. (eds) *Advances in Computational Intelligence Systems. UKCI 2018. Advances in Intelligent Systems and Computing*, vol 840, Springer, Cham, 2019.
- [9] Debashree Devi, Saroj K. Biswas & Biswajit Purkayastha, Learning in presence of class imbalance and class overlapping by using one-class SVM and undersampling technique, *Connection Science*, 2019. DOI: 10.1080/09540091.2018.1560394.
- [10] Mishra S., Chawla M. A Comparative Study of Local Outlier Factor Algorithms for Outliers Detection in Data Streams. In: Abraham A., Dutta P., Mandal J., Bhattacharya A., Dutta S. (eds) *Emerging Technologies in Data Mining and Information Security. Advances in Intelligent Systems and Computing*, vol 813. Springer, Singapore, 2019.
- [11] Huang, Yingfu, and Qiurong Zhang, "Identification of anomaly behavior of ships based on KNN and LOF combination algorithm," *AIP Conference Proceedings*. Vol. 2073. No. 1. AIP Publishing, 2019.
- [12] T. Kanungo, N. S. Netanyahu and A. Y. Wu, "An Efficient k-Means Clustering Algorithm: Analysis and Implementation," *IEEE Transactions On Pattern Analysis And Machine.*, vol. 24, pp. 881-892, 2002.

- [13] Kodinariya, Trupti M., and Prashant R. Makwana. "Review on determining number of Cluster in K-Means Clustering." *International Journal* Vol.1 No.6, pp. 90-95, 2013.
- [14] Sankar Mahadevan, Sirish L. Shah, Fault detection and diagnosis in process data using one-class support vector machines, *Journal of Process Control*, Volume 19, Issue 10, 2009, pp. 1627-1639, ISSN 0959-1524, <https://doi.org/10.1016/j.jprocont.2009.07.011>.
- [15] Breunig, Markus M., *et al.* "LOF: identifying density-based local outliers." *ACM sigmod record*. Vol. 29. No. 2. ACM, 2000.